

Assignment 02

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CSE320: Data Communications

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Q₁

From the figure,

cycle completed = 4

Time to complete = 10 ns

$$\therefore \text{Time period, } T = \frac{10}{4} = 2.5 \text{ ns} = 2.5 \times 10^{-9} \text{ s}$$

$$\therefore \text{Frequency, } f = \frac{1}{T} = \frac{1}{2.5 \times 10^{-9}} = 400 \times 10^6 \text{ Hz}$$

By observing the graph, it is suffering Attenuation.

Q₂

Given,

bandwidth = 20 kHz

Minimum frequency = 25 kHz

Let,

p = Maximum f_q^n

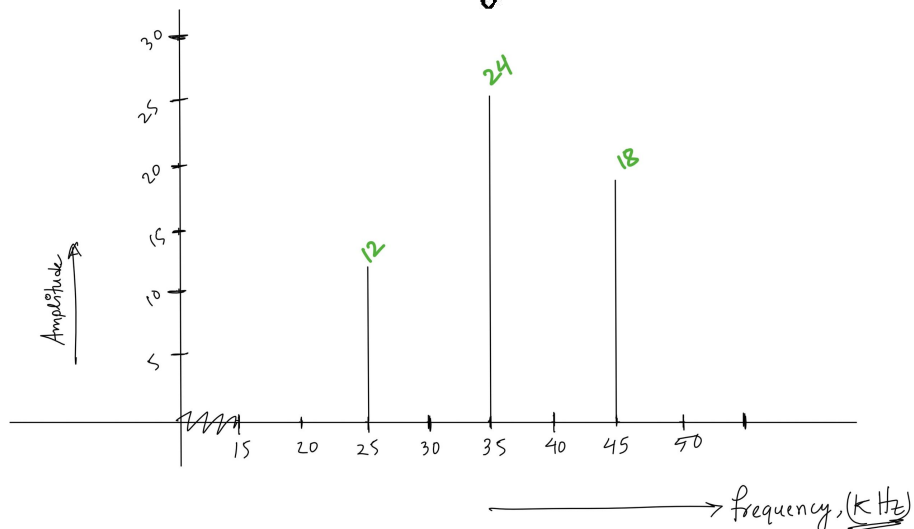
q = Minimum f_q^n

$$\therefore p - q = 20 \text{ kHz}$$

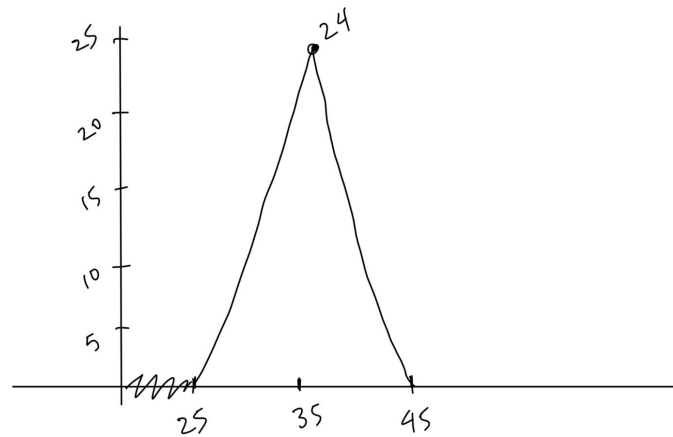
$$\therefore q = 25 \text{ kHz}$$

$$\therefore p = 20 + 25 = 45 \text{ kHz}$$

$$\therefore \text{Mid frequency} = \frac{45 + 25}{2} = \frac{70}{2} = 35 \text{ kHz}$$



Q3



Q4

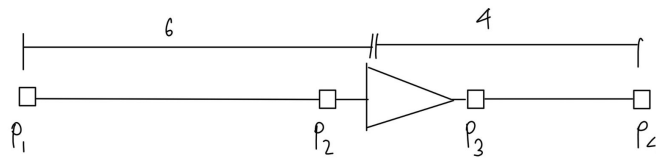
$$\begin{aligned}\text{Total pixel in monitor} &= 1920 \times 1080 \\ &= 2073600 \text{ pixels} \\ &= 2073600 \times 24 \text{ bits} \\ &= 49.7664 \times 10^6 \text{ bits}\end{aligned}$$

$\therefore$ After 5% compression,

$$\begin{aligned}\text{pixel count} &= -(0.05 \times 49.7664 \times 10^6) + 49.7664 \times 10^6 \\ &= -2488320 + 49.7664 \times 10^6 \\ &= 47278080 \text{ bits.}\end{aligned}$$

$$\begin{aligned}\therefore \text{Bitrate} &= 47278080 \times 60 = \cancel{4.97664 \times 10^{361} \text{ b}} \\ &= 2.8366848 \times 10^9 \text{ bps.}\end{aligned}$$

Q5



For 1 km, signal deteriorates = 3 km W

$$\therefore \text{ " 6 " , " " " " } = 6 \times 3 = 18 \text{ km W}$$

$\therefore$ In P_1 , signal strength = 20 W

$$\therefore \text{ " } P_2, \text{ " " " " } = 20 - 18 \text{ W} = 2 \text{ W}$$

$$\begin{aligned} \therefore \text{ Attenuation} &= 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \left(\frac{2}{20} \right) \\ &= -10 \text{ dB} = A_1 \end{aligned}$$

After amplification,

$$\text{signal strength in } P_3 = 15 \times 2 \text{ W} = 30 \text{ W}$$

$$\begin{aligned} \therefore \text{ Attenuation} &= 10 \log_{10} \frac{P_3}{P_2} = 10 \log_{10} \frac{30}{2} \\ &= 11.76 \text{ dB} = A_2 \end{aligned}$$

And,

$$\text{For 4 km, signal deteriorates} = 4 \times 3 = 12 \text{ W}$$

$$\therefore \text{ Signal strength in } P_4 = 30 - 12 = 18 \text{ W}$$

$$\therefore \text{Attenuation} = 10 \log_{10} \frac{P_1}{P_3} = 10 \log_{10} \left(\frac{18}{30} \right) = -2.219 \text{ dB} = A_3$$

$$\therefore \text{Total Attenuation} = A_1 + A_2 + A_3$$

$$= -10 - 2.219 + 11.76 \text{ dB}$$

$$= -0.4581 \text{ dB}$$

Q6

Given,

$$\text{bandwidth} = 100 \text{ kHz} = 100 \times 10^3 \text{ Hz}$$

Data Rate = 75 of Shannon Capacity

$$\text{Number of character} = 1.35 \times 10^6$$

$$= 1.35 \times 10^6 \times 8 \text{ bits}$$

(a)

$$\text{Actual bitrate} = \frac{\text{Total bits}}{\text{Total time}} = \frac{1.35 \times 10^6 \times 8}{5 \times 60} = 3.6 \times 10^6 \text{ bps}$$

$$\therefore \text{Shannon Capacity} = \frac{\text{Actual bitrate}}{0.75} = \frac{3.6 \times 10^6}{0.75} = 4.8 \times 10^6 \text{ bps}$$

$$(b) \text{Shanon Capacity} = \log_2 (1 + \text{SNR}) \times B$$

$$\Rightarrow 4.8 \times 10^6 = \log_2 (1 + \text{SNR}) \times 100 \times 10^3$$

$$\Rightarrow \text{SNR} = 2.814 \times 10^{14}$$

© Given,

$$C = \text{Nyquist Bitrate} = 3.6 \times 10^6 \text{ bps}$$

∴ We know,

$$C = 2B \times \log_2 L$$

$$\Rightarrow 3.6 \times 10^6 = 2 \times 100 \times 10^3 \times \log_2 L$$

$$\Rightarrow \log_2 L = \frac{3.6 \times 10^6}{2 \times 100 \times 10^3}$$

$$\Rightarrow \log_2 L = 18$$

$$\Rightarrow L = 2^{18} = 262144 = \text{Signal Level} \quad \underline{\text{(Ans)}}$$

Q7

$$\text{Queuing Time, } Q = 2 \mu s = 2 \times 10^{-6} \text{ s}$$

$$\text{Processing Time, } P = 1 \mu s = 1 \times 10^{-6} \text{ s}$$

$$\begin{aligned} \text{File size, } S &= 5 \text{ MB} = 5 \times 1024^2 \text{ byte} \\ &= 5 \times 1024^2 \times 8 \text{ bits} \end{aligned}$$

$$\# \text{ Routers} = 10$$

$$\begin{aligned} \text{Distance, } D &= 2000 \text{ Km} \\ &= 2000 \times 10^3 \text{ m} \end{aligned}$$

$$\text{Speed, } S = 2 \times 10^8 \text{ ms}^{-1}$$

$$\begin{aligned} \text{Throughput} &= 5 \text{ Mbps} \\ &= 5 \times 10^6 \text{ bps} \end{aligned}$$

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Speed}} = 0.01 \text{ s}$$

$$\text{Transmission Time} = \frac{\text{Size}}{\text{Bandwidth}} = \frac{5 \times 1024^2 \times 8}{5 \times 10^6} = 8.38608$$

$$\therefore \text{Total Delay} = \text{Propagation Time} + \text{Transmission Time} + \text{Queuing Time} \\ + \text{Processing Time}$$

$$= 0.01 + 8.388608 + 2 \times 10^{-6} + 1 \times 10^{-6}$$

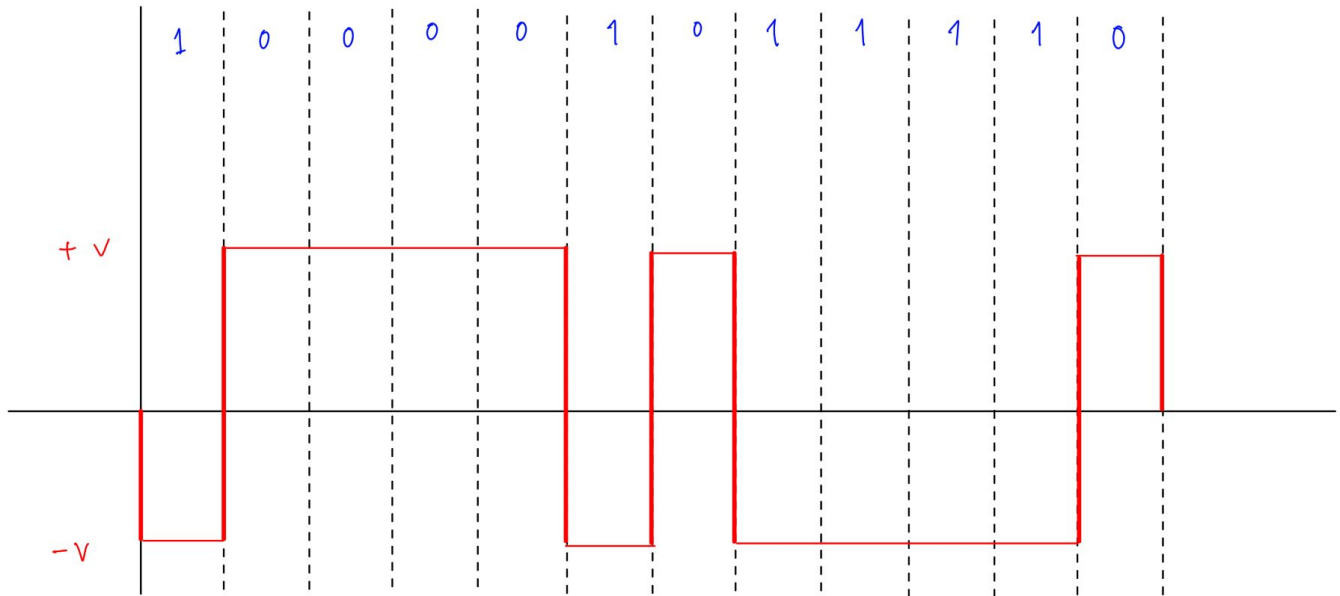
$$= 8.39861 \text{ sec.}$$

→ Dominant → Transmission Time

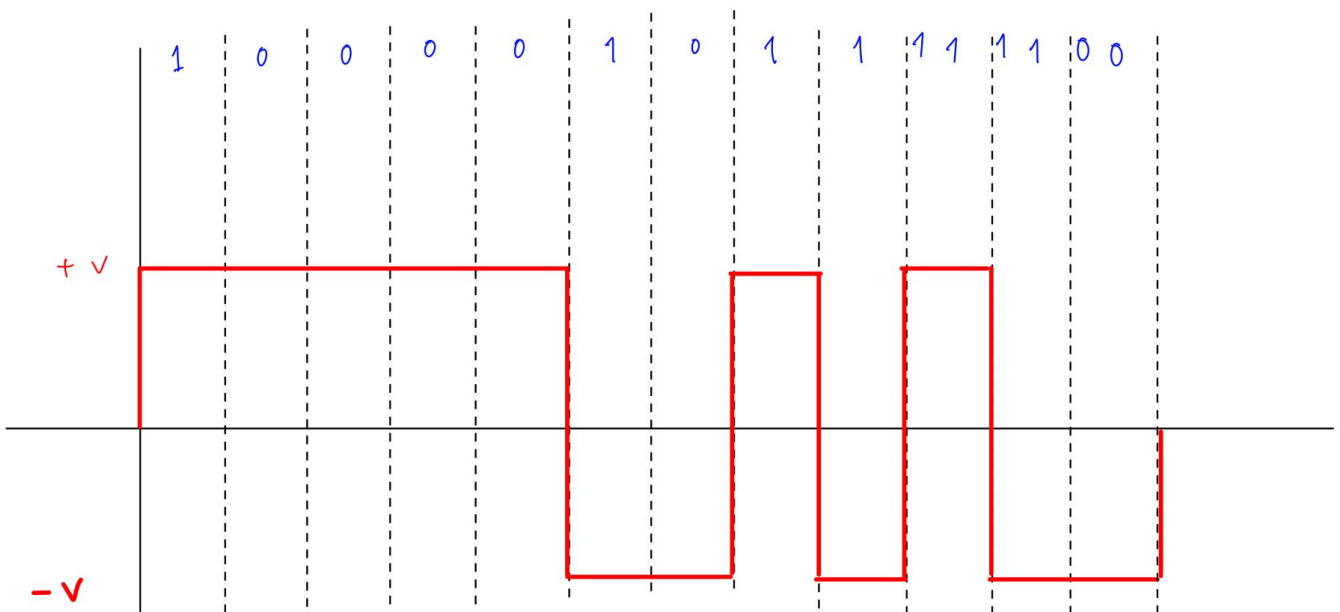
→ Negligible → Queuing Time & Processing Time.

Q8

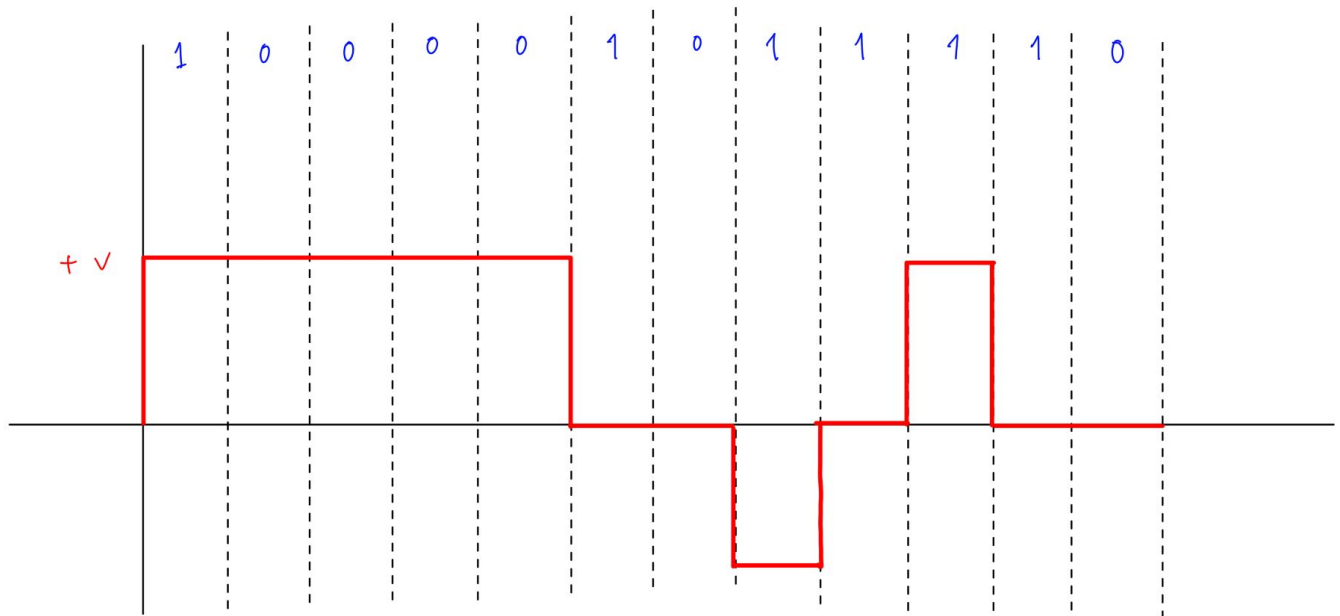
NRZ-L, Assuming $0 \rightarrow \oplus$ ve
 $1 \rightarrow \ominus$ ve



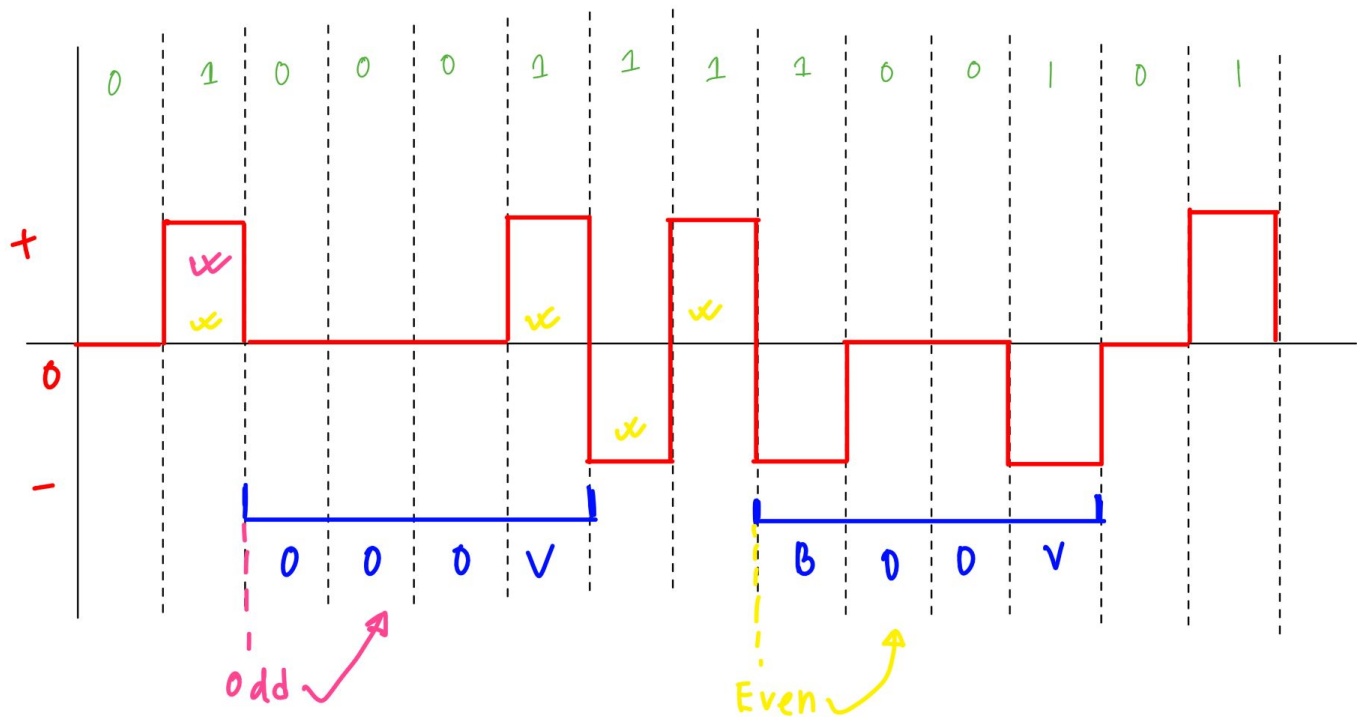
NRZ-I, Assuming $0 \rightarrow$ same
 $1 \rightarrow$ inverse $\rightarrow$ started with positive



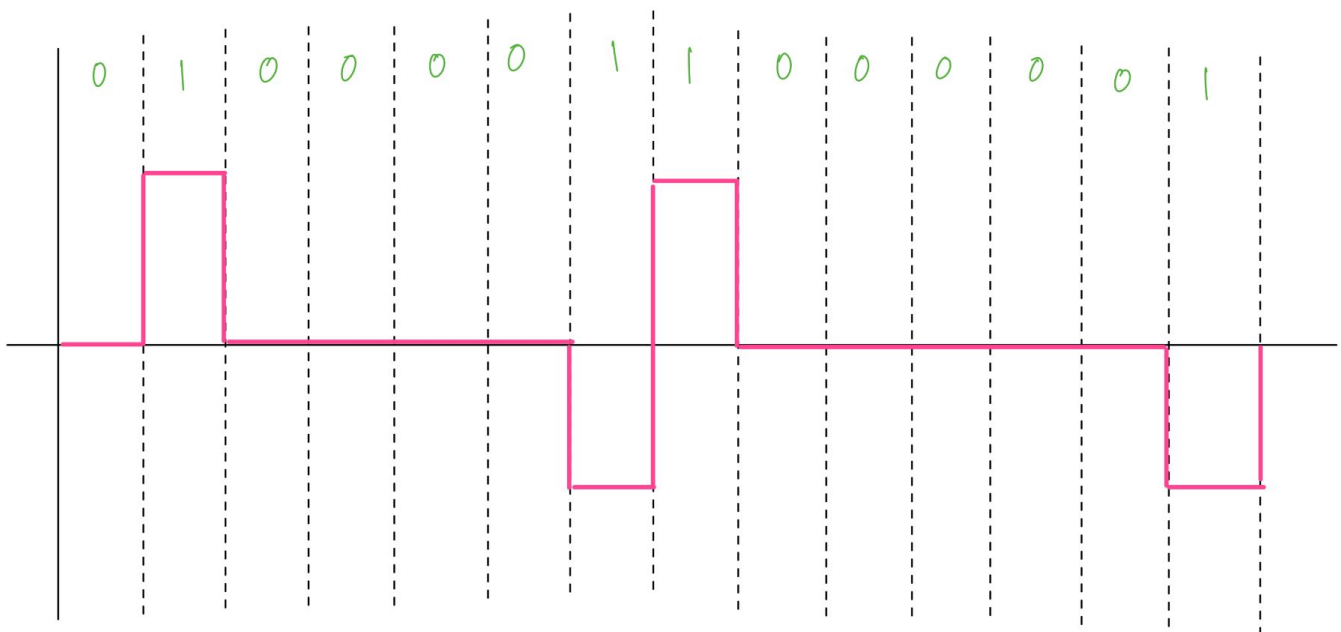
MLT-3



Q-9 Given,



Decoded,



For Line Encoding, Manchester Encoding is comparatively better.

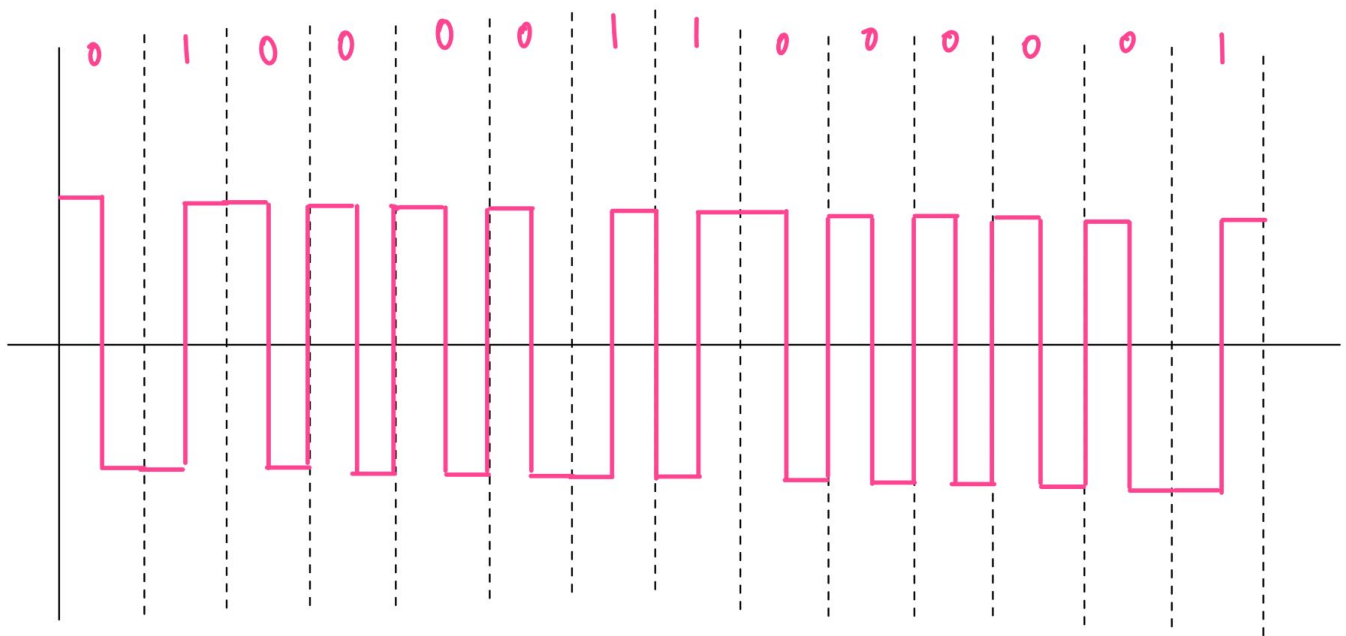
Reasons: ① At least one transition per bit

② No DC component

③ Simple decoding

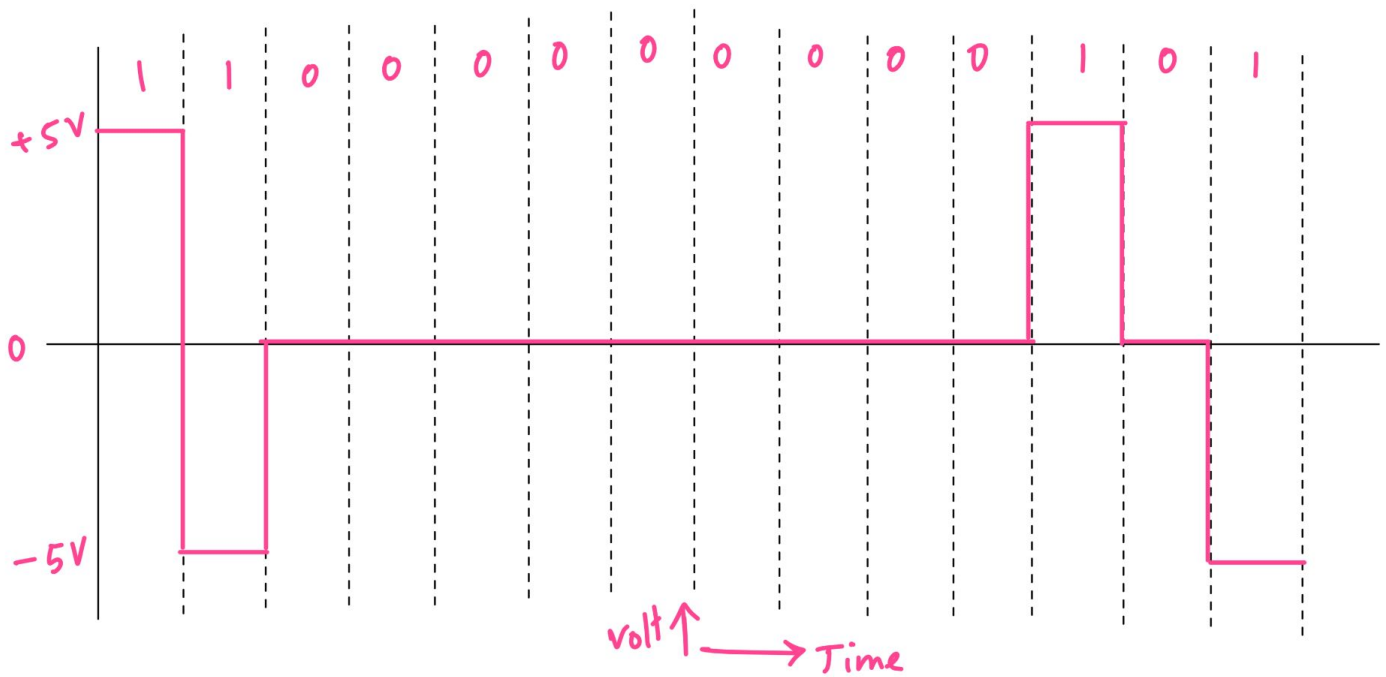
④ No baseline wandering

⑤ No lack of self synchronization.



Q₁₀

① Bipolar AMI was used, where 0 is kept in 0 voltage and for bit 1, alternate/inverse polarity is used.



Decoded: 11000000000101

② Problem: If there is a long sequence of 0, it lacks self synchronization.

③ After observing, we have consecutive 9 zeros. So, B8ZS is must suitable.

